fn conservative_discrete_laplacian_tail_to_alpha

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Proof for conservative_discrete_laplacian_tail_to_alpha.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `scale` is of type `RBig`.
- Argument `tail` is of type `UBig`.

Caller-verified

None

Pseudocode

```
def conservative_discrete_laplacian_tail_to_alpha(scale: RBig, tail: UBig) -> f64:
    check_tail_arguments(scale, RBig(tail))
    numer = f64.inf_cast(-RBig(tail) / scale).inf_exp()
    denom = f64.neg_inf_cast(RBig.ONE / scale).neg_inf_exp().neg_inf_add(1.0)
    return numer.inf_div(denom)
```

Postcondition

Definition 1.1. Define $X \sim \mathcal{L}_{\mathbb{Z}}(0, s)$, a random variable following the discrete laplace distribution:

$$\forall x \in \mathbb{Z} \qquad \Pr[X = x] = \frac{1 - e^{-1/s}}{1 + e^{-1/s}} e^{-|x|/s} \tag{1}$$

Lemma 1.2. Assume $X \sim \mathcal{L}_{\mathbb{Z}}(0, s)$, and $t \geq 0$, where $s =$ `scale` and $t =$ `tail`.

$$\alpha = P[X > t] = \frac{e^{-t/s}}{e^{1/s} + 1} \tag{2}$$

Proof.

$$
\begin{aligned}
\alpha &= P[X > t] \\
&= \sum_{x=t+1}^{\infty} \frac{1 - e^{-1/s}}{1 + e^{-1/s}} e^{-|x|/s} \\
&= \frac{1 - e^{-1/s}}{1 + e^{-1/s}} \sum_{x=t+1}^{\infty} e^{-x/s} && \text{since } t \geq 0 \\
&= \frac{1 - e^{-1/s}}{1 + e^{-1/s}} \frac{e^{-(t+1)/s}}{1 - e^{-1/s}} && \text{since } \sum_{x=t}^{\infty} p^x = \frac{p^t}{1-p} \text{ if } |p| < 1, \text{ let } p = e^{-1/s} \\
&= \frac{e^{-(t+1)/s}}{1 + e^{-1/s}} \\
&= \frac{e^{-t/s}}{e^{1/s} + 1}
\end{aligned}
$$

□

Theorem 1.3. For any inputs `scale` and `tail` satisfying the preconditions, `conservative_discrete_laplacian_tail_to_alpha` either:

- returns `Err(e)`, or
- returns `Ok(out)` with `out` $\geq \Pr[X >$ `tail`$]$, where $X \sim \mathcal{L}_{\mathbb{Z}}(0,$ `scale`$)$.

Proof. Let $s =$ `scale` and $t =$ `tail`. `check_tail_arguments` returns `Err(e)` unless $s > 0$, while $t \geq 0$ holds by type `UBig`. Otherwise, if no remaining operation returns `Err(e)`, `numer` computes $e^{-t/s}$ with conservative upward rounding, so `numer` $\geq e^{-t/s} > 0$, and `denom` computes $e^{1/s}+1$ with conservative downward rounding, so $0 <$ `denom` $\leq e^{1/s} + 1$. Dividing `numer` by `denom` with conservative upward rounding therefore yields a result at least $\frac{e^{-t/s}}{e^{1/s}+1}$, which equals $\Pr[X > t]$ by Lemma 1.2. □